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SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement ID: SIH26059**
- **Problem Statement Title: AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System.**
- **Theme: Transportation & Logistics**
- **PS Category: Software**
- **Team ID:**
- **Team Name: LimitLess**

❖ Proposed Solution

- **Integrated Intelligence:** Antarctic Traverse integrates satellite, sea-ice, ocean, and weather data to forecast ice conditions, detect icebergs, and predict their movement.

- **From Prediction to Action:** These forecasts are transformed into a dynamic hazard and risk map to enable safer navigation for Antarctic research vessels.

- **Unified Decision Support:** AI-based forecasting, physics-informed iceberg trajectory modelling, and explainable, risk-aware routing are integrated into a unified GIS-based decision-support workflow.

Outcome: One Unified view of hazards, route risks and safer alternatives.

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TECHNICAL APPROACH

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- ## Technology Stack
  Python • PyTorch / TensorFlow • XGBoost • GeoPandas / Rasterio • QGIS/Leaflet • FastAPI

- ## Core Models & Algorithms
  U-Net • ConvLSTM • Optical Flow • Physics-informed ML with residual correction • A*/Dijkstra • NSGA-II

Data Sources	Prediction	Risk Assessment	Route Optimization	Decision Dashboard
Sentinel-1	Sea-Ice Concentration	Risk Index	Dynamic Graph	GIS-based Map
NSIDC/AMSR2	Ice Berg	Hazard Score	A*/Dijkstra	Safe Route
ERA5 • GLORYS	Trajectory Forescasting	Uncertainty	NSGA-II	Alternatives

Method: Ingest → Align → Predict → Quantify Risk → Optimize → Visualize

FEASIBILITY AND VIABILITY

- **Feasibility:** Public datasets and reproducible models make a practical, modular MVP feasible.
- **Key Risks:** Sparse Antarctic ground truth, ice-edge bias, satellite data latency, and computational constraints.
- **Mitigation:** Phase the development: start with U-Net/ConvLSTM, then add uncertainty-aware outputs and physics-informed drift modelling; validate against BYU/NIC, SCAR, and labelled Sentinel-1 data.

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IMPACT AND BENEFITS

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- **Safer Antarctic Voyages:** Forecasted sea-ice conditions, iceberg detection, and trajectory prediction enable safer route decisions for research vessels.
- **Better Operational Decisions:** A unified GIS dashboard brings forecast layers, hazard/risk scores, confidence and ranked route alternatives into one operator-facing view.
- **Fuel & Environmental Efficiency:** Multi-objective routing balances risk, travel time, and fuel consumption, enabling operators to compare safer, more efficient route alternatives.

RESEARCH AND REFERENCES

1. **Braakmann-Folgmann et al. (2023):** *Mapping the extent of giant Antarctic icebergs with deep learning*

2. **Andersson et al. (2021):** *Seasonal Arctic sea ice forecasting with probabilistic deep learning*

3. **Wang et al. (2023):** *Subseasonal Prediction of Regional Antarctic Sea Ice by a Deep Learning Model*

4. **Hui et al. (2017):** *Satellite-Based Sea Ice Navigation for Prydz Bay, East Antarctica*

5. **Liu et al. (2025):** *Vessel Safety Navigation Under the Influence of Antarctic Sea Ice*

6. **Budge & Long (2018):** *A Comprehensive Database for Antarctic Iceberg Tracking Using Scatterometer Data*